

Pathology-Level ML Model Performance Summary (Test Set)
Classical ML models trained on neural network pathology predictions

Neural Network Mode	Pathology	ML Algorithm	AUC Mean	AUC Std	AUC 95% CI	F1 Mean	F1 Std	Optimal Threshold	Threshold Std	Samples	Positive Rate	Cross-Val Folds
HMV-MIL	A-lines	Random Forest	0.758	0.026	[0.726-0.790]	0.678	0.040	0.466	0.035	43689	0.472	5
HMV-MIL	A-lines	Logistic Regression	0.795	0.014	[0.778-0.813]	0.728	0.010	0.402	0.026	43689	0.472	5
HMV-MIL	A-lines	XGBoost	0.794	0.013	[0.779-0.810]	0.712	0.020	0.442	0.008	43689	0.472	5
HMV-MIL	Large Consolidation	Random Forest	0.863	0.013	[0.847-0.880]	0.511	0.064	0.456	0.030	43689	0.110	5
HMV-MIL	Large Consolidation	Logistic Regression	0.919	0.004	[0.913-0.924]	0.562	0.060	0.754	0.043	43689	0.110	5
HMV-MIL	Large Consolidation	XGBoost	0.901	0.012	[0.886-0.915]	0.588	0.048	0.382	0.024	43689	0.110	5
HMV-MIL	Pleural Effusion	Random Forest	0.674	0.058	[0.602-0.745]	0.112	0.050	0.260	0.049	43689	0.011	5
HMV-MIL	Pleural Effusion	Logistic Regression	0.808	0.039	[0.759-0.856]	0.086	0.033	0.748	0.031	43689	0.011	5
HMV-MIL	Pleural Effusion	XGBoost	0.829	0.041	[0.778-0.880]	0.109	0.035	0.232	0.029	43689	0.011	5
HMV-MIL B-lines and Small Consolidation	B-lines and Small Consolidation	Random Forest	0.644	0.024	[0.614-0.674]	0.529	0.048	0.444	0.009	43689	0.405	5
HMV-MIL B-lines and Small Consolidation		Logistic Regression	0.690	0.022	[0.662-0.717]	0.612	0.018	0.396	0.015	43689	0.405	5
HMV-MIL B-lines and Small Consolidation		XGBoost	0.684	0.018	[0.662-0.707]	0.580	0.035	0.414	0.013	43689	0.405	5
NoKeyframe	A-lines	Random Forest	0.754	0.064	[0.675-0.833]	0.681	0.048	0.464	0.023	43689	0.472	5
NoKeyframe	A-lines	Logistic Regression	0.790	0.036	[0.745-0.835]	0.725	0.031	0.434	0.021	43689	0.472	5
NoKeyframe	A-lines	XGBoost	0.790	0.054	[0.723-0.856]	0.711	0.041	0.432	0.015	43689	0.472	5
NoKeyframe	Large Consolidation	Random Forest	0.829	0.114	[0.687-0.971]	0.429	0.135	0.398	0.036	43689	0.110	5
NoKeyframe	Large Consolidation	Logistic Regression	0.908	0.030	[0.872-0.945]	0.463	0.243	0.680	0.065	43689	0.110	5
NoKeyframe	Large Consolidation	XGBoost	0.862	0.117	[0.716-1.007]	0.469	0.139	0.382	0.036	43689	0.110	5
NoKeyframe	Pleural Effusion	Random Forest	0.678	0.064	[0.599-0.757]	0.113	0.048	0.256	0.038	43689	0.011	5
NoKeyframe	Pleural Effusion	Logistic Regression	0.730	0.065	[0.648-0.811]	0.065	0.030	0.634	0.078	43689	0.011	5
NoKeyframe	Pleural Effusion	XGBoost	0.819	0.066	[0.737-0.900]	0.109	0.051	0.228	0.023	43689	0.011	5
NoKeyframe B-lines and Small Consolidation	B-lines and Small Consolidation	Random Forest	0.645	0.032	[0.606-0.685]	0.520	0.079	0.440	0.010	43689	0.405	5
NoKeyframe B-lines and Small Consolidation		Logistic Regression	0.672	0.031	[0.634-0.711]	0.480	0.267	0.386	0.021	43689	0.405	5
NoKeyframe B-lines and Small Consolidation		XGBoost	0.686	0.046	[0.629-0.743]	0.556	0.121	0.406	0.011	43689	0.405	5